

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4				Indicative content: The main parts of the electromagnetic spectrum are gamma rays, X-rays, ultraviolet, visible light, infra-red, microwaves and radio waves. Images of the universe are taken by Earth based systems, and space craft and transmitted to Earth. Electromagnetic waves are used to study structures in the universe e.g. • X-ray images of stars • ultra violet images of the Sun, galaxies and the planets • visible light images of the Sun, the planets, moons and galaxies • infra-red images of the Sun, the planets and the Milky Way • microwave images of the Sun • radio wave images of stars • gamma images of black holes 5–6 marks Full list of em spectrum regions, 3 or more types of images, from Earth or space craft. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i>	6	1		1		1

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
				3–4 mark Lists 4+ regions of the em spectrum and states 2 examples of the types of images that can be produced. OR Full list (7) regions of the em spectrum with one use with Earth-based or space craft – based. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1–2 marks Lists 1+ regions of the em spectrum OR One region and its use <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
				Question 4 total	6			6		